Analytics & Big Data

Session 5: Regression 2

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🎯 Learning objectives

- **Explain** how business problems can be formulated as binary classification tasks and modeled using logistic regression.
- **Understand** how logistic regression uses a linear predictor and sigmoid function to produce probabilities, apply thresholds, and interpret log-odds and coefficients.
- **Evaluate** classification models using confusion matrices and metrics such as accuracy, precision, recall, and F1 score.
- **Apply** model predictions to decision-making by selecting appropriate thresholds based on business costs and expected value.

Business problem: Churn prediction

Predicting customer churn

A lost customer is often a predictable customer. Firms collect many traces of customer relationships — usage, spending, support interactions, contract data. The analytical challenge is to turn these traces into an early warning system.

Typical variables in a churn dataset

- **Who is the customer?** tenure, segment, contract type, subscription tier
- **How do they use the service?** login frequency, feature usage, inactivity days
- **How satisfied are they?** complaints, support tickets, response times, NPS/satisfaction
- **Are there warning signs?** payment delays, downgrades, cancellations of add-ons, price sensitivity

Prediction goal

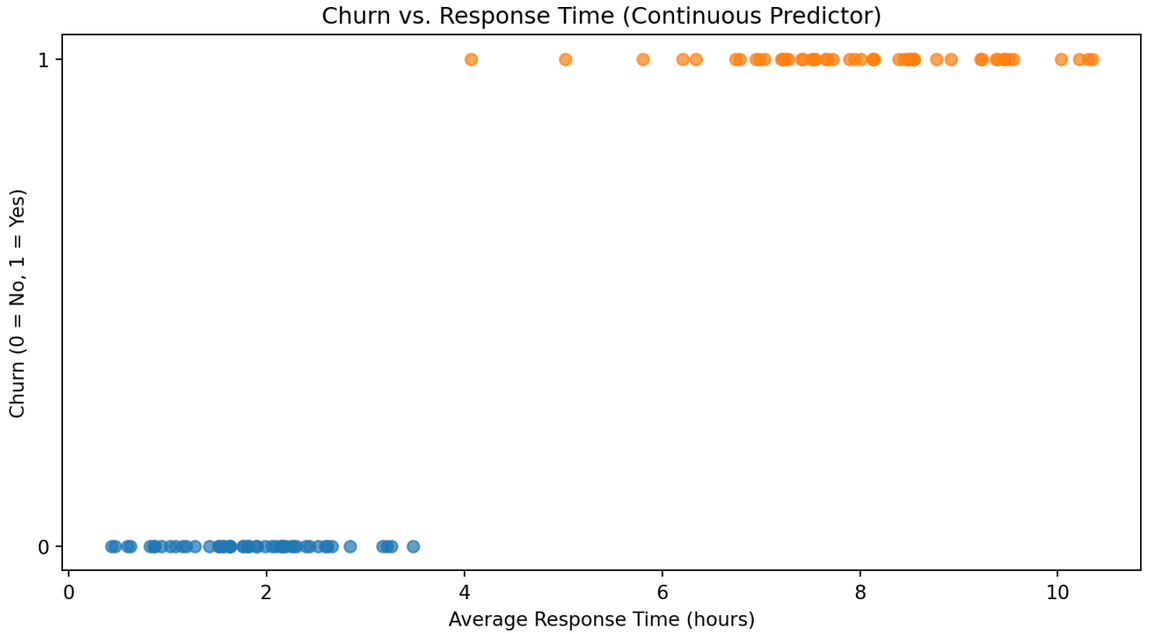
Predict:

$$P(\text{churn} = 1 \mid X)$$

where X contains the observed customer characteristics and behaviors.

Geometric view of the problem

Churn prediction is a **binary classification problem**:



Possible approach: We could use Linear Regression with a threshold at `0.5` to classify output:

- If $f(x) < 0.5$, predict `y=0`
- If $f(x) \geq 0.5$, predict `y=1`

Logistic regression

Classification

Examples:

- Online transactions: Fraudulent (yes/no)?
- Customer churn: yes/no?
- EMail: Spam/not spam?

$$y \in \{0, 1\}$$

with

0: “Negative class” (e.g., no fraud, no churn, no spam)
1: “Positive class” (e.g., fraud, churn, spam)

Logistic regression model

Ideally, our model should predict the **probability** of $y_i = 1$. This would allow us to apply a threshold for classification, but it would also give us more information (how likely an observation belongs to the *positive class*).

It means we need a model that produces predicted probabilities between 0 and 1:

Step 1: Linear model

$$z_i = \beta_0 + \beta_1 x_i$$

Step 2: Transform to probability via sigmoid function

$$\mathbb{P}(y_i = 1 \mid x_i) = \sigma(z_i) \text{ with } \sigma(z) = \frac{1}{1 + e^{-z}}$$

Logistic regression model

$$\mathbb{P}(y_i = 1 \mid x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

Key intuition:

- Linear predictor $z \rightarrow$ transformed by the sigmoid (logistic) function
- Output always in $[0, 1]$
- Can be interpreted as a probability (next slide)

Log-odds

Logistic regression assumes that the **log-odds of the outcome are linear in the predictors**:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_i$$

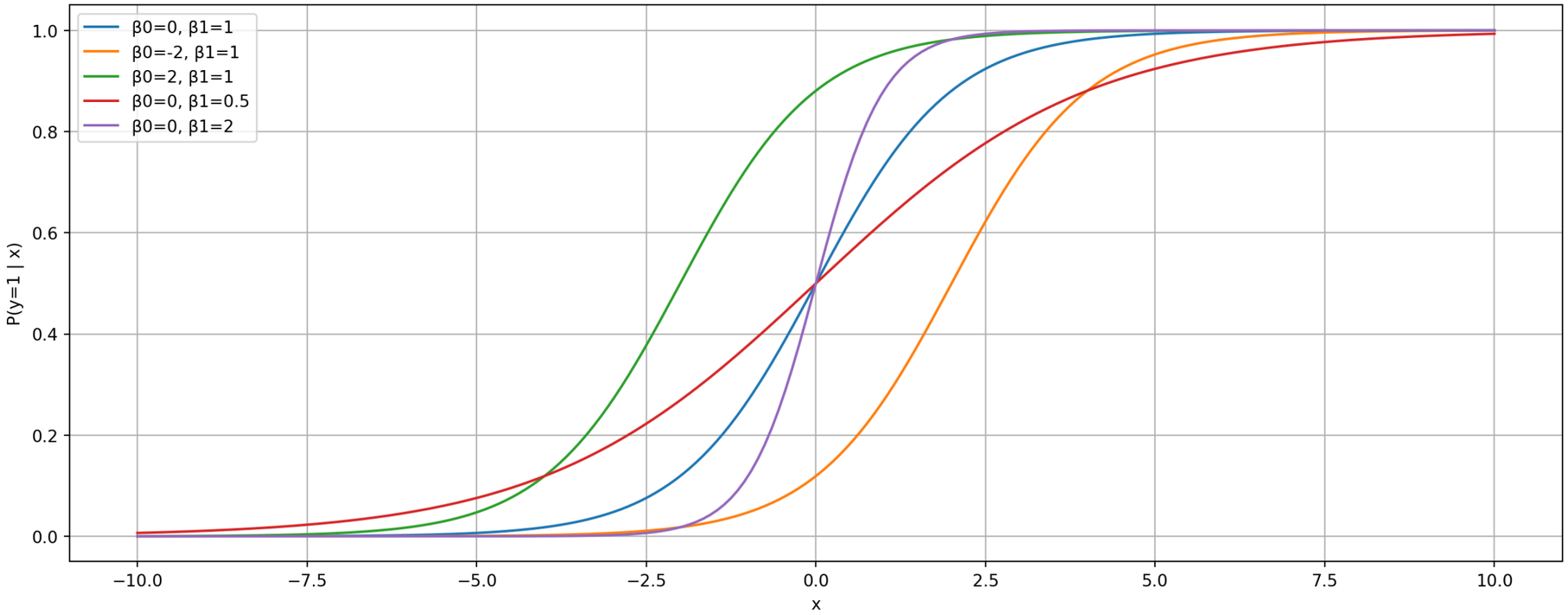
Note: $\frac{p_i}{1-p_i}$ *are the "odds".*

This implies a **sigmoid-shaped relationship** between x_i and the probability of $y_i = 1$:

$$p_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

The sigmoid function serves as a *link function* to convert the linear predictor into a *probability*.

The Sigmoid function



Note

- Changing β_0 **shifts** the curve left/right
- Changing β_1 **changes slope** (flatter vs. steeper transition)

Estimation: Maximum likelihood

Instead of minimizing squared errors, logistic regression chooses parameters that make the observed data most likely.

Core idea: We want predicted probabilities to match observed outcomes:

- If $y_i = 1$ → predicted probability should be **high**
- If $y_i = 0$ → predicted probability should be **low**

Likelihood function

For each observation: $P(y_i \mid x_i) = p_i^{y_i}(1 - p_i)^{1-y_i}$

For all observations:

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i}(1 - p_i)^{1-y_i}$$

Maximization: Choose β_0, β_1 to maximize this likelihood

💡 **Intuition:** We search for the model that assigns high probability to what actually happened.

No need to memorize the formula. Understand the idea.

Logistic regression in Python

We estimate the model using the sklearn library:

```python
import pandas as pd
from sklearn.linear_model import LogisticRegression

df = pd.DataFrame({
    "response_time": [1, 2, 3, 5, 7, 8, 9, 11],
    "churn":         [0, 0, 0, 0, 1, 1, 1, 1]
})
X = df[["response_time"]]
y = df["churn"]

model = LogisticRegression()
model.fit(X, y)

print("Intercept (β0):", model.intercept_[0])
print("Slope (β1):", model.coef_[0][0])
```

```
Intercept (β0): -6.201807152628621
Slope (β1): 1.0606214874843112
```

Interpretation of coefficients

Logistic regression coefficients are not directly effects on probability. They describe effects on the log-odds.

Coefficient meaning

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

- β_1 : change in **log-odds** when x increases by 1

More intuitive: odds ratios

Exponentiate the coefficient: e^{β_1}

- Factor change in the **odds**
- Example:
  - $e^{\beta_1} = 1.5$ → odds increase by **50%**
  - $e^{\beta_1} = 0.7$ → odds decrease by **30%**

Important implication

- Effect on probability is **nonlinear**
- Depends on current level of x

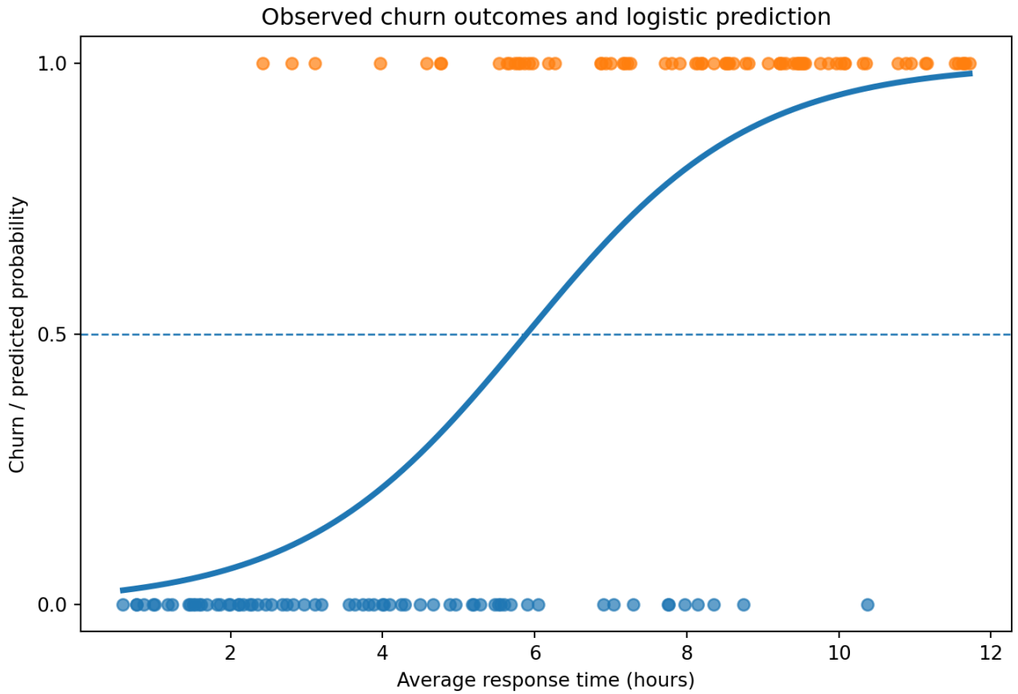
Evaluation

Predicted probabilities

A logistic regression model predicts a **probability** of churn:

$$P(\text{churn}_i = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

> A threshold (such as **0.5**) is needed to convert probabilities into class labels.
>
> Comparing predicted vs. actual class labels allows us to evaluate the performance of a classifier like logistic regression.



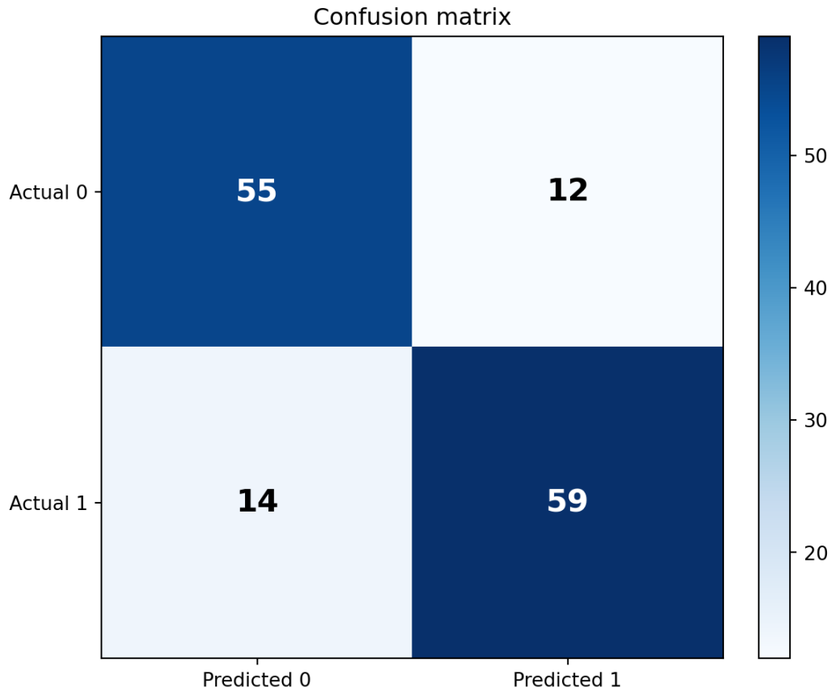
Note: there are no meaningful *residuals* (or R^2), as in linear regression models.

Confusion matrix

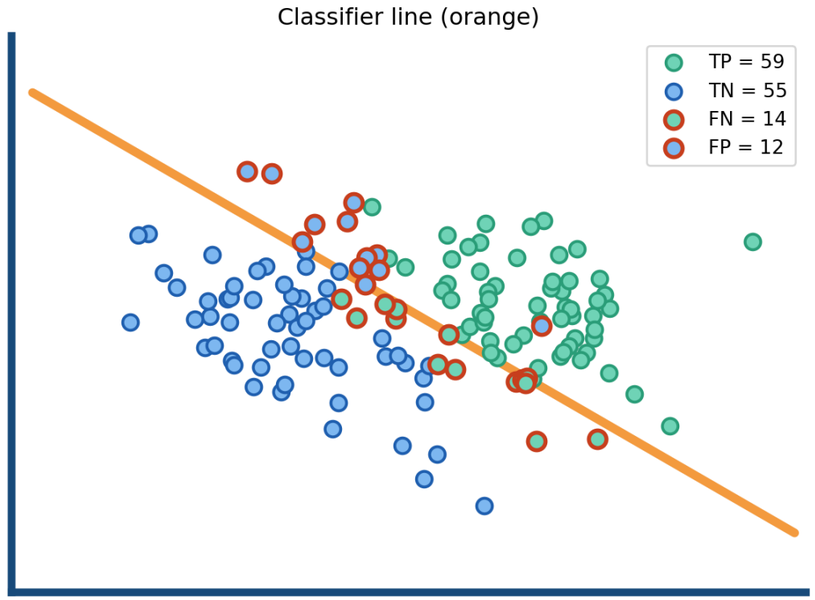
Using a **0.5 threshold**, predicted probabilities are turned into classes.

The confusion matrix summarizes:

- **True negatives (TN):** predicted no churn, observed no churn
- **False positives (FP):** predicted churn, observed no churn
- **False negatives (FN):** predicted no churn, observed churn
- **True positives (TP):** predicted churn, observed churn



Evaluation metrics



Metric	Formula	Value	Interpretation
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	81.4%	Overall share of correct predictions
Precision	$\frac{TP}{TP+FP}$	83.1%	Among predicted positives, how many are actually positive?
Recall	$\frac{TP}{TP+FN}$	80.8%	Among actual positives, how many did the model identify?
F1	$2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	81.9%	Balance between precision and recall

ROC curve and AUC

ROC (Receiver Operating Characteristic) curve summarizes the trade-off between:

- **True Positive Rate (aka. recall or sensitivity)**
- **False Positive Rate**

As we vary the **classification threshold**, the model becomes more or less conservative in predicting `churn = 1`.

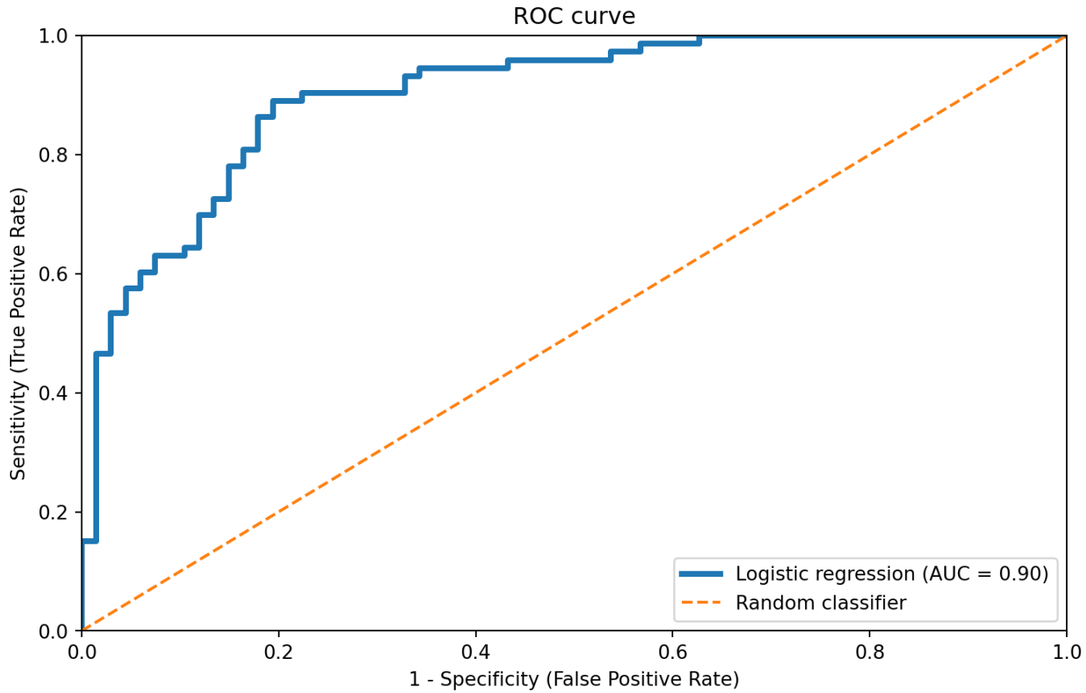
Area under the curve (AUC) summarizes ROC performance in a single number:

- **AUC = 0.5**: no better than random guessing
- **AUC = 1.0**: perfect separation
- **Higher AUC**: better ranking of churners vs. non-churners

A good classifier achieves:

- **high true positive rate**
- **low false positive rate**

→ curves closer to the **top-left corner** indicate better model fit.



From prediction to decisions

From prediction to decision

A logistic regression model predicts a **probability of churn**:

$$p(x) = P(\text{churn} = 1 \mid x)$$

This probability is based on historical data, reflecting how similar customers behaved in the past, and capturing a baseline churn risk. It is not a certainty.

Suppose a firm has two possible actions:

- **Intervene**: offer a retention incentive
- **Do nothing**

So the business problem is no longer only:

> Who is likely to churn?

It becomes:

> For whom is an intervention worth it?

Decision-making under uncertainty

We can use the expected value criterion to **translate predictions into decisions** by combining:

- **Predicted churn probability** from the model
- **Business consequences** of the action

A simple payoff structure:

Action	Customer would churn	Customer would stay
Intervene	+20	-10
Do nothing	-100	0

Interpretation:

- Saving a likely churner creates value
- Intervening with a customer who would stay creates unnecessary cost
- Missing a true churner is very costly

Expected value of (not) intervening

Action	Customer would churn	Customer would stay
Intervene	+20	-10
Do nothing	-100	0

Probabilities:

- With probability $p(x)$, the customer is at risk and intervention helps
- With probability $1 - p(x)$, the customer would stay anyway and intervention wastes resources

If we **intervene**, the expected value is:

$$EV(intervene) = p(x) \cdot 20 + (1 - p(x)) \cdot (-10) = 30p(x) - 10$$

If we **do nothing**, the expected value is:

$$EV(do\_nothing) = p(x) \cdot (-100) + (1 - p(x)) \cdot 0 = -100p(x)$$

Based on Elkan (2001).

Optimal decision rule

Instead of selecting one action, we should intervene whenever:

$$EV(intervene) > EV(do\_nothing)$$

Insert the expected values:

$$p(x) \cdot 20 + (1 - p(x)) \cdot (-10) > -100p(x)$$

Simplify:

$$20p(x) - 10 + 10p(x) > -100p(x)$$

$$130p(x) > 10$$

$$p(x) > 0.077$$

Decision threshold

This gives a clear rule:

> Intervene if $p(x) > 0.077$ (if predicted churn probability exceeds 7.7%).

The threshold is low because:

- Losing a customer is **very costly**
- Intervening is **relatively cheap**

So even a small churn probability can justify action.

💡 A classification threshold does not have to be 0.5. It should depend on the **business context**.

Two sources of knowledge

This decision combines two complementary inputs:

1. Data-driven component

- Logistic regression estimates:

$$p(x) = P(\text{churn} = 1 \mid x)$$

- Learned from historical data
- Provides a probability (what is likely to happen)

2. Business-driven component

- A threshold that determines when to act
- Ideally derived from expected value considerations
- Based on business costs, strategy, and domain knowledge

> The model produces probabilities — not decisions.
>
> There are multiple ways to set a threshold for classification, depending on the context.
>
> A decision only emerges when probabilities are combined with a chosen threshold — this is where predictive analytics becomes prescriptive analytics.

Outlook: Machine learning

From statistical models to machine learning

Logistic regression is a classification model and represents a basic form of machine learning:

- learns from data
- makes predictions
- but with strong structural assumptions

It is:

- Interpretable
- Fast and robust
- Often a baseline model

Logistic regression is often the first model to try — simple, transparent, and surprisingly effective.

💡 More complex models are useful only if they improve performance meaningfully

From statistical models to machine learning

New challenges in machine learning

1. Overfitting

- Models may fit noise instead of signal
- Good training performance ≠ good generalization

2. Role of data

- More observations and features become critical

3. Evaluation

- Same metrics (accuracy, precision, recall, F1)
- But evaluated on unseen data

4. Model tuning

- Not only thresholds
- Additional hyperparameters

Logistic regression	Machine learning models
Linear decision boundary	Can learn nonlinear boundaries
Strong assumptions	Fewer assumptions
Selected variables	Many variables (high-dimensional)
Interpretable	Often less interpretable

Key shift:

- From structured, theory-driven models
- To flexible, data-driven models

Summary

- Logistic regression is a method for **binary classification**, where the goal is to predict whether an outcome occurs (e.g., churn vs. no churn).
- It predicts **probabilities** $P(y = 1 \mid X)$ using a linear predictor combined with a **sigmoid function**, ensuring outputs between 0 and 1.
- Coefficients are interpreted in terms of **log-odds** (or odds ratios), not direct probability changes.
- Parameters are estimated via **maximum likelihood**, aiming to match predicted probabilities with observed outcomes.
- Model performance is evaluated using **classification metrics** such as confusion matrices, accuracy, precision, recall, F1, and ROC/AUC.
- A **threshold** is required to convert probabilities into class labels, and should be chosen based on **business costs**, not fixed at 0.5.

Survey: Session 5

https://forms.gle/mVWw3z7ftFn48gDY6

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏

References

Elkan, C. (2001). The foundations of cost-sensitive learning. *International Joint Conference on Artificial Intelligence*, *17*, 973–978. https://doi.org/10.5555/1642194.1642224